

Advanced Machine Learning

DATA 642

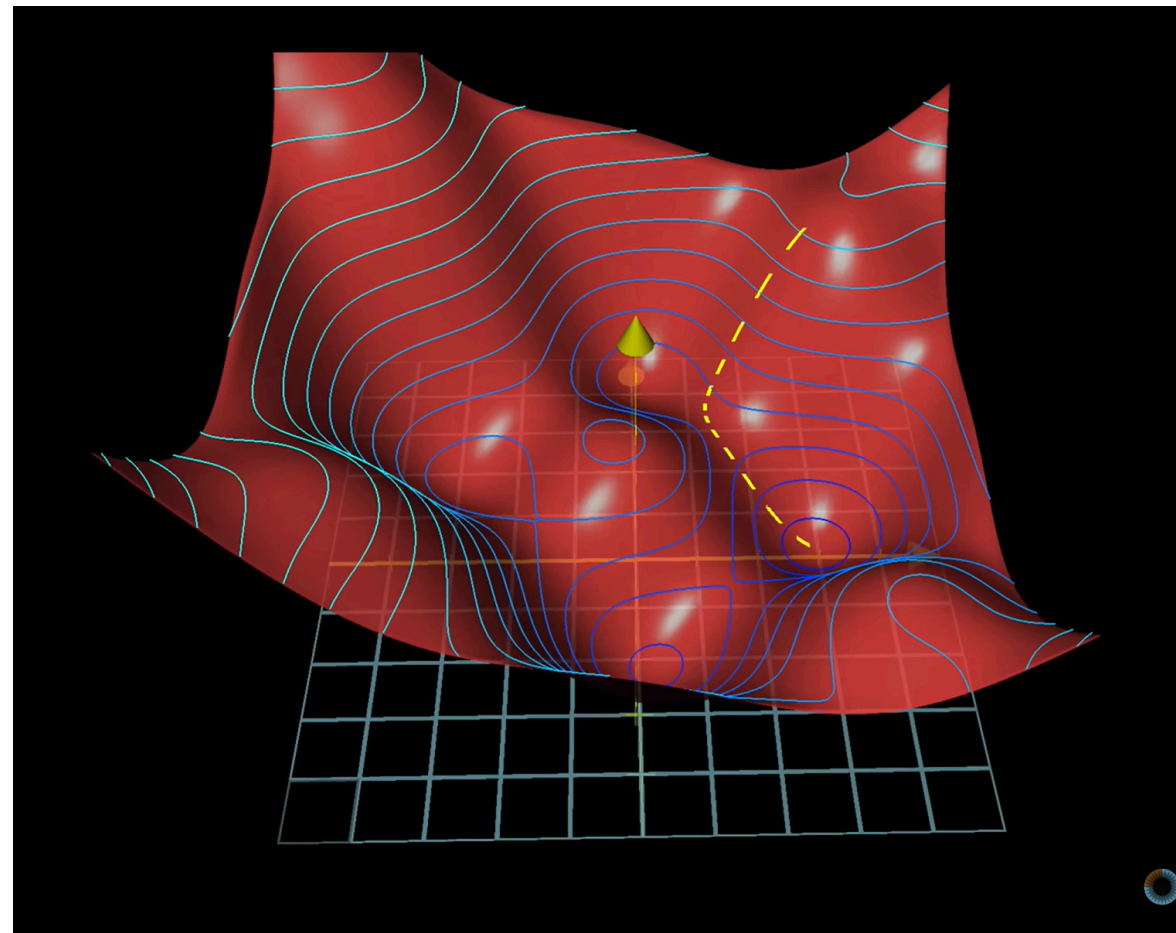
Lecture 12 — Intuitive Walkthrough

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly, and 3blue1brown, or 3b1b for short, which is a YouTube channel about discovery and creativity in math, with an emphasis on visualizations. <https://www.3blue1brown.com/topics/neural-networks>

Optimization is at the heart of training

What we really do is to find what is the set of weight and biases that minimize a loss function

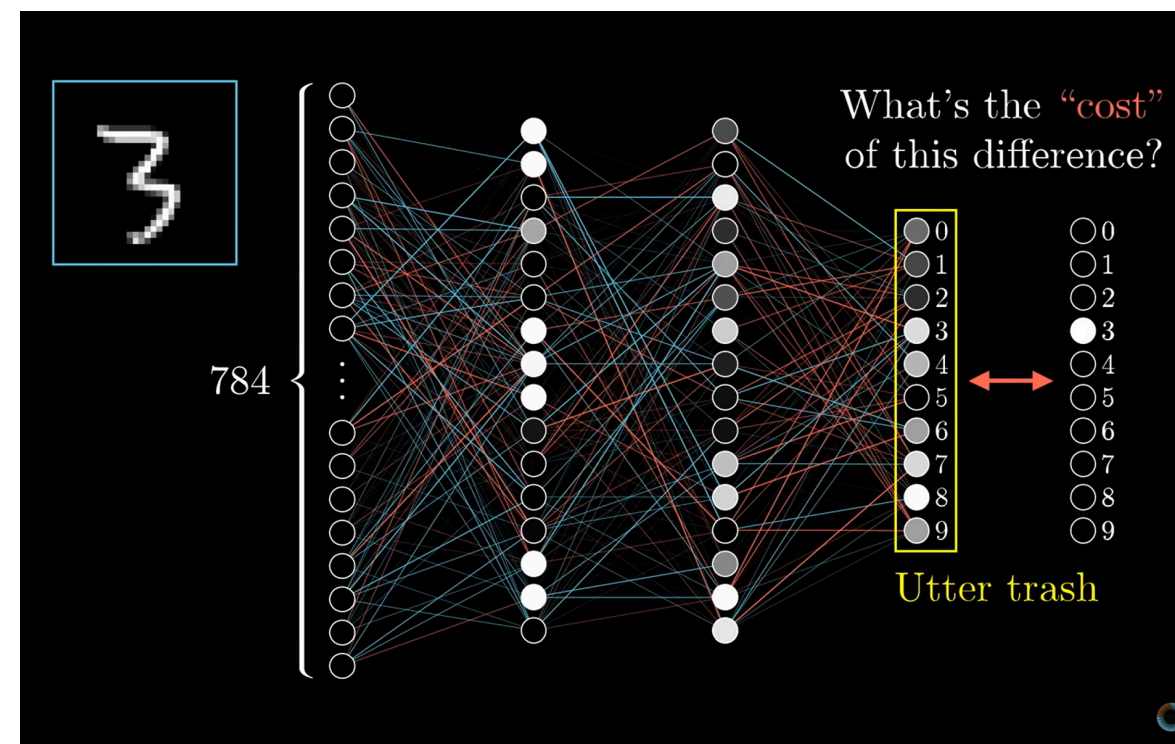
- The loss function quantifies the error between predicted outputs and actual targets.
- It captures the discrepancy between predicted and ground truth values.



Training Multilayer Neural Networks: An Intuitive Overview

Forward Propagation:

- Input data passes through the network layer by layer.
- Each layer performs a linear transformation followed by an activation function.
- Output of one layer becomes input to the next until the output layer produces predictions.

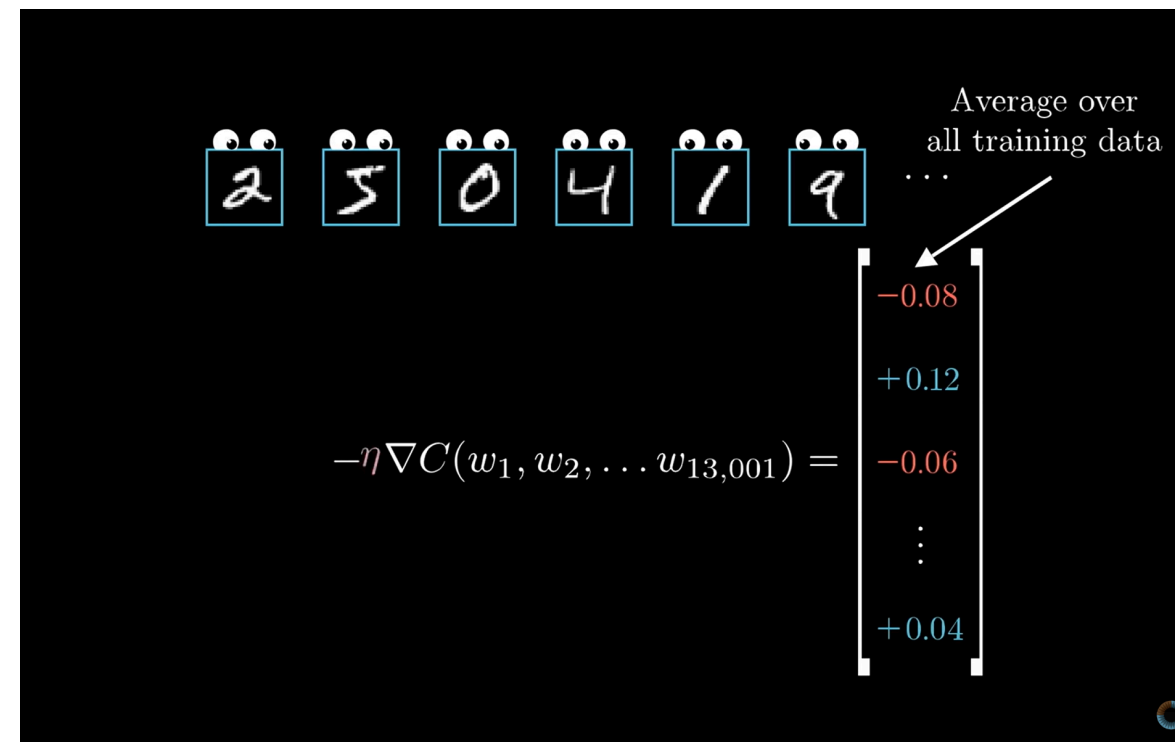


Loss Calculation:

Loss between predicted outputs and actual targets is calculated using a loss function that quantifies how well the model is performing on the training data.

Backpropagation:

- Gradients of the loss function with respect to the weights are computed using the chain rule of calculus.
- These gradients are used to adjust the weights in a direction that reduces the loss.



Gradient Descent Optimization:

- Gradient descent algorithms (e.g., SGD, Adam) are used to update weights efficiently.
- Learning rate and weight updates are adjusted iteratively to minimize the loss function.

Iterative Training:

- Forward propagation, loss calculation, backpropagation, and weight updates are repeated iteratively for a fixed number of epochs or until convergence.
- Model learns to extract relevant features and adjust parameters to make better predictions.

Evaluation:

- Trained model is evaluated on a separate validation dataset to assess performance.
- Metrics such as accuracy, precision, recall, and F1-score are commonly used for evaluation.